

FLISA II, Norway

WMO: 013920

Lat: 60.6139N

Lon: 12.0122E

Elev: 607

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-2.5	-11.6	3.0	-7.0	-6.9	3.9	-2.6	14.2	34.3	13.0	33.7	2.0	30	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	80.8	61.5	77.1	60.9	73.2	59.1	64.3	75.4	62.7	72.4	61.1	69.9	4.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	80.7	66.9	59.0	76.2	65.6	57.4	71.9	64.3	29.7	75.4	28.5	72.5	27.4	70.2	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.0	9.6	-11.4	84.5	6.0	3.5	-15.7	87.0	-19.2	89.1	-22.6	91.1	-27.0	93.6		
(5)			-12.2	66.9	6.0	2.1	-16.5	68.4	-20.1	69.6	-23.4	70.8	-27.8	72.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	21.4	23.2	30.9	41.1	50.5	57.7	62.6	59.1	51.9	40.9	32.1	23.6		
	DBStd	16.41	10.86	10.10	8.75	6.12	7.41	5.60	5.33	4.38	5.56	6.89	8.18	12.48		
	HDD50	4282	885	750	591	275	85	4	0	1	44	291	537	819		
	HDD65	8689	1350	1170	1056	716	453	228	115	189	394	747	987	1284		
	CDD50	1127	0	0	0	9	100	237	390	282	100	9	0	0		
	CDD65	59	0	0	0	0	3	11	40	5	0	0	0	0		
	CDH74	680	0	0	0	0	60	163	415	42	0	0	0	0		
	CDH80	121	0	0	0	0	3	27	90	1	0	0	0	0		
(14) Wind	WSAvg	4.2	3.8	3.8	4.3	4.6	4.7	4.7	4.2	4.0	4.1	4.1	3.9	3.8		
(15) Precipitation	PrecAvg	27.6	1.8	1.3	1.3	1.7	2.4	3.0	3.2	3.5	2.7	2.7	2.3	2.0		
	PrecMax	34.7	4.0	2.7	3.5	4.0	6.5	5.4	6.4	6.4	5.6	5.9	4.3	3.5		
	PrecMin	21.9	0.2	0.4	0.1	0.2	0.4	0.2	0.6	1.4	0.7	0.5	0.7	0.2		
	PrecStd	3.6	1.0	0.6	0.7	0.9	1.5	1.3	1.5	1.2	1.1	1.2	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	44.2	57.1	67.4	79.9	83.5	86.5	78.4	71.0	60.3	51.1	47.5		
		MCWB	40.1	38.1	45.2	50.8	58.7	61.0	64.4	61.9	58.1	53.3	48.0	44.3		
	2%	DB	39.5	40.4	50.8	63.1	74.8	79.1	82.6	74.5	67.1	56.4	48.4	44.2		
		MCWB	37.2	36.8	41.5	49.3	56.6	59.6	63.4	60.4	56.6	51.0	45.5	41.7		
	5%	DB	37.4	38.0	45.8	58.4	70.2	73.8	78.9	71.5	64.2	53.8	45.7	41.6		
		MCWB	35.4	35.4	38.3	46.6	55.0	58.1	62.6	59.4	55.2	49.3	43.4	39.1		
	10%	DB	35.0	35.7	42.4	53.7	64.8	69.9	74.8	68.3	61.2	51.2	43.2	38.8		
		MCWB	33.1	33.7	36.7	43.7	53.2	56.0	60.9	58.1	54.3	47.7	41.3	36.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	40.1	45.8	52.7	61.7	63.7	67.3	64.8	61.3	55.5	48.8	44.9		
		MCDB	42.1	42.5	55.2	63.4	72.4	74.1	79.1	71.6	66.0	58.4	50.6	47.0		
	2%	WB	37.8	37.5	42.1	50.2	59.0	61.6	65.4	63.0	58.8	52.3	45.9	42.0		
		MCDB	39.2	39.6	49.3	61.0	70.7	73.7	77.7	70.2	64.0	55.3	47.9	44.1		
	5%	WB	35.5	35.6	39.6	47.7	56.4	59.6	63.7	61.4	57.0	50.1	43.7	39.4		
		MCDB	37.1	37.5	44.5	57.5	67.0	71.3	75.8	68.1	62.3	52.9	45.5	41.1		
	10%	WB	33.4	33.8	37.4	44.7	53.8	57.5	62.1	59.9	55.3	48.1	41.4	36.6		
		MCDB	34.9	35.7	41.7	52.4	63.3	67.8	72.3	66.1	60.0	51.0	43.1	38.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	11.7	16.6	18.7	18.5	18.0	17.8	16.4	16.1	11.9	9.1	10.2		
		MCDBR	9.8	12.2	21.0	26.8	26.9	25.5	24.9	21.5	19.9	14.1	10.8	11.0		
		MCWBR	8.1	9.3	13.0	14.7	12.3	11.1	10.1	10.1	11.3	9.7	9.6	10.0		
		MCWBR	8.7	10.8	19.3	24.3	22.1	22.2	21.8	17.4	16.1	11.4	10.1	10.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.242	0.287	0.314	0.354	0.356	0.348	0.368	0.359	0.327	0.313	0.262	N/A		
		taud	2.190	2.279	2.317	2.347	2.425	2.470	2.430	2.450	2.527	2.479	2.278	N/A		
		Ebn at Noon	168	226	258	267	276	280	270	264	255	218	159	N/A		
		Edh at Noon	12	20	27	32	32	31	32	29	23	17	11	N/A		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	87	293	763	1216	1501	1706	1527	1196	790	352	112	53		
		RadStd	14	55	82	149	150	127	208	115	96	69	32	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page